

Multiplicative Inverse, Fermat's little Theorem

Robert Y. Lewis

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Overview

- 1 Multiplicative Inverses (8.6.1)
- 2 Cancellation (8.6.2)
- 3 Fermat's Little Theorem (8.6.3)

Back to basics

Definition: The *multiplicative inverse* of a number x is a number x^{-1} such that:
 $x \cdot x^{-1} = 1$.

Division by x is really multiplication by x^{-1} .

Over the reals, what values have inverses? Everybody but zero.

Over the integers, what values have inverses? Only 1 and -1 .

Over the integers mod n , what values have inverses?

Example, mod 10

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	5	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

What specific values have inverses? 1, 3, 7, 9.

What specific values do *not* have inverses? 0, 2, 4, 5, 6, 8.

General rule? a has an inverse (mod n) iff $\gcd(a, n) = 1$.

Back to solving

$$3x + 4 \equiv 27 \pmod{11}$$

$$3x \equiv 23 \pmod{11} \quad \text{add } -4 \text{ to both sides}$$

Want to multiply both sides by $3^{-1} = 4$, since $3 \times 4 \equiv 1 \pmod{11}$.

$$3x \equiv 23 \pmod{11}$$

$$4 \times 3x \equiv 4 \times 23 \pmod{11} \quad \text{multiply both sides by 4}$$

$$12x \equiv 92 \pmod{11} \quad \text{simplify}$$

$$x \equiv 4 \pmod{11} \quad \text{congruence}$$

Double check: plug in 4 or 15 to the original formula.

Inverse mod prime

General rule for existence of multiplicative inverses? a has an inverse mod n iff $\gcd(a, n) = 1$.

If this rule holds, all values (except zero!) have inverses mod a prime.

Lemma: If p is prime and k is not a multiple of p , then k has a multiplicative inverse modulo p .

Proof: Since p is prime and k is not a multiple of p , $\gcd(p, k) = 1$. Therefore, there are s and t such that $1 = sp + tk$. So, mod p , that's $1 \equiv tk$, or $t \equiv k^{-1} \pmod{p}$. QED.

Example: What's the multiplicative inverse of 3 (mod 11)?

$$\text{gcdcombo}(3, 11) = (4, -1, 1)$$

So? 4 works. Because $1 = 4 \times 3 - 1 \cdot 11$, so, mod 11, that's $1 \equiv 4 \times 3$.

Back to dividing both sides

Earlier, we saw:

$$7 \equiv 28 \pmod{3}$$

$$1 \equiv 4 \pmod{3} \quad \text{divide by 7}$$

Doesn't actually work, in general:

$$12 \equiv 6 \pmod{3}$$

$$4 \not\equiv 2 \pmod{3} \quad \text{divide by 3}$$

Why? Because we're really talking about multiplying both sides by 0^{-1} , which doesn't exist.

Cancellation

Theorem.

If we have

$$ak \equiv bk \pmod{p}$$

and p is prime and $k \not\equiv 0 \pmod{p}$, then $a \equiv b \pmod{p}$.

Proof. $k^{-1} \pmod{p}$ exists. So, multiply both sides by k^{-1} and congruence is maintained.

Permuting

Corollary: Suppose p is prime and k is not a multiple of p . Then, the sequence of remainders on division by p of the sequence:

$$1 \cdot k, 2 \cdot k, \dots, (p-1) \cdot k$$

is a permutation of the sequence:

$$1, 2, \dots, (p-1).$$

Example, $k = 3, p = 11$:

i	1	2	3	4	5	6	7	8	9	10
$\times k$	3	6	9	12	15	18	21	24	27	30
$\text{mod } p$	3	6	9	1	4	7	10	2	5	8

Permutation proof

Proof: The sequence of remainders contains $p - 1$ numbers. Since $i \times k$ is not divisible by p (neither contains a factor of p) for $i = 1, \dots, p - 1$, all these remainders are in $[1, p)$ by the definition of remainder.

Claim: if $i \cdot k \equiv j \cdot k \pmod{p}$, then $i = j$. (Cancel k ; since $1 \leq i < p$, $i \pmod{p} = i$, same for j .)

So, $i - 1$ distinct values between 1 and $i - 1$: it's a permutation.

It's a magic shuffle function. Useful for randomization and sending secret messages!

Fermat's little theorem

Theorem: Suppose p is prime and k is not a multiple of p . Then:

$$k^{p-1} \equiv 1 \pmod{p}.$$

$$(p-1)!$$

$$= 1 \cdot 2 \cdot \dots \cdot (p-1)$$

Proof:	$= \text{rem}(k, p) \cdot \text{rem}(2k, p) \cdots \text{rem}((p-1)k, p)$	Defn. of factorial
	$\equiv k \cdot 2k \cdots (p-1)k \pmod{p}$	Permutation lemma
	$\equiv (p-1)!k^{p-1} \pmod{p}$	Congruence of mult. algebra

Note that $(p-1)!$ is not a multiple of p because none of $1, 2, \dots, (p-1)$ contain a factor of p . So, by the Cancellation lemma, we can cancel $(p-1)!$ from the top and bottom, proving the claim. QED

Inverses from Fermat's little theorem

Since $k^{p-1} \equiv 1 \pmod{p}$ and $k^{p-1} = k \cdot k^{p-2}$, that tells us that k^{p-2} is the multiplicative inverse for k .

We can compute $k^{p-2} \pmod{p}$ efficiently using a technique called exponentiation by repeated squaring.

Running time is $2 \log p$, just like “gcdcombo”.

Exponentiation by Repeated Squaring Idea

Can always compute a^k by $k - 1$ multiplications of a .

If k is even, can compute it with $k/2 - 1$ multiplications of a to get $a^{k/2}$. Then, $a^k = (a^{k/2})^2$. So, one more multiplication and we're there.

If k is odd, similar trick to get $a^{(k-1)/2}$, then square, then multiply one more a .

Repeating this idea, the number of multiplications is on the order of $2 \log k$.

Exponentiation by Repeated Squaring

```
def repsq(a,k):
    if k == 0: return(1)
    if k % 2 == 0:
        sqroot = repsq(a,k/2)
        return(sqroot*sqroot)
    sqrootdiva = repsq(a,(k-1)/2)
    return(sqrootdiva*sqrootdiva*a)
```

(Note: we're using the infix notation % to mean “remainder,” often read out loud as “mod.”)

Never need to multiply big numbers

When doing multiplication mod n , we can always mod n the numbers first.

Example:

$$\begin{aligned} 7415 \times 2993 \% 3 \\ = 22193095 \% 3 \\ = 1 \end{aligned}$$

OR:

$$\begin{aligned} (7415 \% 3) \times (2993 \% 3) \% 3 \\ (2 \% 2) \% 3 \\ = 1. \end{aligned}$$

Proof

Theorem. $ab \% n = (a \% n)(b \% n) \% n$.

Proof.

$$a = q_1n + r_1$$

$$b = q_2n + r_2$$

$$ab = (q_1n + r_1)(q_2n + r_2)$$

$$ab = (q_1q_2n + q_1r_2 + q_2r_1)n + r_1r_2$$

So, cancelling out the multiple of n , $ab \% n = r_1r_2 \% n$.

Exponentiation by Repeated Squaring Mod Style

```
def repsqmodn(a,k,n):  
    a := a % n  
    if k == 0: return(1)  
    if k % 2 == 0:  
        sqrootdiva = repsqmodn(a,k/2,n)  
        return((sqrootdiva*sqrootdiva) % n)  
    sqrootdiva = repsqmodn(a,(k-1)/2,n)  
    return((sqrootdiva*sqrootdiva*a) % n)
```